BandLink subsystem ownership

The bracelet's subsystems for 490A and the hand-offs between them. Each subsystem gets one accountable owner and one reviewer; the owner of the subsystem that produces data writes the spec for that hand-off. Arrows point from producer to user.

Owners and reviewers are blank on purpose: assign them at the team meeting, then fill in the table below.

SUBSYSTEMS AND HAND-OFFS · OWNERS TO BE ASSIGNED AT THE TEAM MEETING

490B scope

Positioning

Owner: ___ Rev: ___

MAX-M10S GNSS fix, PPS capture
BNO085 heading (rotation vector)
Navigation: distance, bearing, arrow angle

SW-002 · SW-006

own position into packet

Radio link

Owner: ___ Rev: ___

SX1262 LoRa 915 MHz, GPS-timed slots
Packet format, AES-128-CCM, dedupe
Range testing; relay (Stretch 2)

SW-ARCH-03 · SW-001 · SW-004 · TEST-001

group key after tap

Pairing + ranging

Owner: ___ Rev: ___

PN7150 NFC tap-to-pair
Group key + device ID exchange, key store
DW3000 UWB distance (Stretch 2)

SW-003

friends' positions

heading, distance, bearing

Power

Owner: ___ Rev: ___

Power modes: Standby / Typical / Finding
Battery, charging, AMOLED VBAT supply
PPK2 current measurements

SW-ARCH-05 · HW-07 · TEST-002

sleep + mode hooks

Firmware platform

Owner: ___ Rev: ___

FreeRTOS tasks, queues, ISR rules
BSP + Nucleo pin map, CubeMX config
Driver interface spec

SW-ARCH-02 · HW-05 · SW-ARCH-04

ui task + UI queue

Display + UI

Owner: ___ Rev: ___

Osptek 1.73" AMOLED (CO5300), CST820 touch
LVGL screens: finder, group, pairing
Haptics (ERM now, LRA in 490B)

SW-ARCH-06 · UI-001 · SW-005

measured currents

docs + pin map in repo

Integration + parts

Owner: ___ Rev: ___

490A parts order, toolchain setup
Portable MVP units, end-to-end demo
Proposal + BOM kept in sync

HW-04 · SW-000 · INT-001 · DOC-03

Repo + process

Owner: ___ Rev: ___

Git workflow, branch protection, PR reviews
Decisions log, task plan, architecture docs
Hub board kept in step with the repo

MISC-006 · DOC-04

Later work

decide in 490B

Rev A PCB schematic + layout
Enclosure + strap (3D print)
iPhone app + BLE bridge (OPT-07)

490B cards

Assignment worksheet

SUBSYSTEM	OWNER	REVIEWER	PARTS	REQUIREMENTS	HUB CARDS	BRANCH PREFIX
Positioning			MAX-M10S, BNO085	R-02, R-07	`SW-002, SW-006`	`fw/`
Radio link			SX1262 + 915 MHz antenna	R-01, R-06	`SW-ARCH-03, SW-001, SW-004, TEST-001`	`fw/, test/, docs/`
Pairing + ranging			PN7150; DW3000 (S2)	R-04; R-09 (S2)	`SW-003`	`fw/`
Power			Battery, charger, regulators, PPK2	R-03, R-08	`SW-ARCH-05, HW-07, TEST-002`	`docs/, test/`
Firmware platform			NUCLEO-U5A5ZJ-Q, STM32U5G9 (final)	(all, indirectly)	`SW-ARCH-02, HW-05, SW-ARCH-04`	`docs/, hw/`
Display + UI			Osptek AM173Q466466FLS, ERM / LRA	R-05	`SW-ARCH-06, UI-001, SW-005`	`ui/`
Integration + parts			490A order	MVP demo	`HW-04, SW-000, INT-001, DOC-03`	`hw/, fw/, docs/`
Repo + process			—	—	`MISC-006, DOC-04`	`docs/`
Rev A PCB, enclosure, iPhone app			490B	—	`490B cards`	`hw/, app/`